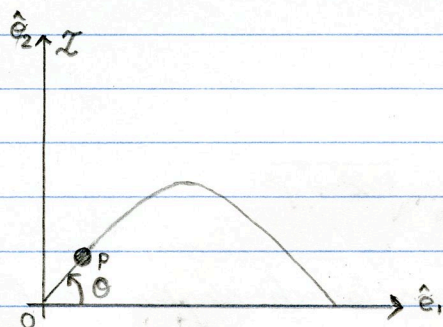




Problem Statement:

- Find the launch angle(s) required to hit a target two meters away with cannon ball with a launch velocity of 10 ms^{-1}
- Find the minimum velocity and the corresponding angle required to hit the same target

Assumptions:

- Quadratic Drag Model
- $g = 10 \text{ ms}^{-2}$
- Cannon ball modelled as a point mass

Kinematics:

$$\begin{aligned}\vec{r}_{P/O} &= x\hat{e}_1 + y\hat{e}_2 \\ \vec{v}_{P/O} &= \dot{x}\hat{e}_1 + \dot{y}\hat{e}_2 ; \hat{e}_v = \frac{\dot{x}\hat{e}_1 + \dot{y}\hat{e}_2}{\sqrt{\dot{x}^2 + \dot{y}^2}} \\ \vec{a}_{P/O} &= \ddot{x}\hat{e}_1 + \ddot{y}\hat{e}_2\end{aligned}$$

Dynamics:

$$\begin{aligned}\Sigma \vec{F}_{P/O} &= m_P \vec{a}_{P/O} \\ \Rightarrow -mg\hat{e}_2 - C_D |\vec{v}_{P/O}|^2 \hat{e}_v &= m(\ddot{x}\hat{e}_1 + \ddot{y}\hat{e}_2) \\ \Rightarrow -mg\hat{e}_2 - C_D (\dot{x}^2 + \dot{y}^2) \left\{ \frac{\dot{x}\hat{e}_1 + \dot{y}\hat{e}_2}{\sqrt{\dot{x}^2 + \dot{y}^2}} \right\} &= m(\ddot{x}\hat{e}_1 + \ddot{y}\hat{e}_2)\end{aligned}$$

$\{ \} \cdot \hat{e}_1$

$$\begin{aligned}-C_D (\dot{x}^2 + \dot{y}^2)^{\frac{1}{2}} \dot{x} &= m\ddot{x} \\ \therefore \ddot{x} &= -\frac{C_D \dot{x}}{m} \sqrt{\dot{x}^2 + \dot{y}^2}\end{aligned}$$

$\{ \} \cdot \hat{e}_2$

$$-mg - C_D (\dot{x}^2 + \dot{y}^2)^{\frac{1}{2}} \dot{y} = m\ddot{y}$$

$$\therefore \ddot{y} = -g - \frac{C_D \dot{y}}{m} \sqrt{\dot{x}^2 + \dot{y}^2}$$

- a. For a specific velocity, there are two launch angles which allow the cannon ball to reach the same target.

Using `FSOLVE`, it is possible to find both these angles by varying the initial guess and thereby redefining the search space. The two required angles corresponding to a launch velocity of 10 m/s are 61.5175° and 13.1865° .

See attached plots and code.

- b. To find the minimum required launch velocity, `FMINCON` was used, where the objective function to be minimized was the distance between the final position of the ball and the target.

See attached plots and code.

The minimum velocity is: 7.718 m/s and the associated angle of launch is 50.144° .

```

% MAE 4730 HW 6.22
% Author: Amlan Sinha, Cornell University

clear all
close all
clc

set(0,'DefaultFigureWindowStyle','docked')

% First launch angle

% System parameters
p.m = 1;
p.g = 10;
p.cd = 0.5;

% Initialization
v0 = 10; % Initial velocity
th0 = pi/3; %Initial launch angle

% Populating the initial state vector
Y0 = [0; v0*cos(th0); 0; v0*sin(th0)];

% Time span
tSpan = linspace(0,10,10^3);

% Optimization: Find an appropriate launch angle
% Objective: x(tf) = 2 m
options = optimoptions('fsolve','stepTolerance',1e-12,'FunctionTolerance',1e-12);
th = fsolve(@code_22ra,th0,options);

% Integration
opts.relTol = 1e-12;
opts.absTol = 1e-12;
opts.Events = @zeroHeight;

% Re-initialize with optimized parameters
Y0_opt = [0; v0*cos(th); 0; v0*sin(th)];

% Integration
[T,Y] = ode45(@code_22a,tSpan,Y0_opt,opts,p);

% Graphics

```



```

figure(1)
plot(Y(:,1),Y(:,3),'k',Y(1,1),Y(1,3),'k*',Y(end,1),Y(end,3)
,'ko')
title('Cannon Ball w/ Quadratic Drag');
xlabel('x(m)');
ylabel('y(m)');

fprintf('The required angle is: %3.3e rad \n',th);

% MAE 4730 HW 6.22
% Author: Amlan Sinha, Cornell University

clear all
close all
clc

set(0,'DefaultFigureWindowStyle','docked')

% Second launch angle

% System parameters
p.m = 1;
p.g = 10;
p.cd = 0.5;

% Initialization
v0 = 10; % Initial velocity
th0 = pi/6; %Initial launch angle

% Populating the initial state vector
Y0 = [0; v0*cos(th0); 0; v0*sin(th0)];

% Time span
tSpan = linspace(0,10,10^3);

% Optimization: Find an appropriate launch angle
% Objective: x(tf) = 2 m
options = optimoptions('fsolve','stepTolerance',1e-
12,'FunctionTolerance',1e-12);
th = fsolve(@code_22ra,th0,options);

% Integration
opts.relTol = 1e-12;
opts.absTol = 1e-12;

```

```

opts.Events = @zeroHeight;

% Re-initialize with optimized parameters
Y0_opt = [0; v0*cos(th); 0; v0*sin(th)];

% Integration
[T,Y] = ode45(@ode_22a,tSpan,Y0_opt,opts,p);

% Graphics
figure(1)
plot(Y(:,1),Y(:,3),'k',Y(1,1),Y(1,3),'k*',Y(end,1),Y(end,3)
,'ko')
title('Cannon Ball w/ Quadratic Drag');
xlabel('x(m)');
ylabel('y(m)');

fprintf('The required angle is: %3.3e rad \n',th);

function dydt = ode_22a(t,y,p)

vmag = sqrt(y(2)^2+y(4)^2);

dydt(1) = y(2);
dydt(2) = -p.cd/p.m*y(2)*vmag;
dydt(3) = y(4);
dydt(4) = -p.g-p.cd/p.m*y(4)*vmag;

dydt = dydt';

end

```

```

% MAE 4730 HW 6.21
% Author: Amlan Sinha, Cornell University

clear all
close all
clc

set(0,'DefaultFigureWindowStyle','docked')

% System parameters
p.m = 1;
p.g = 10;
p.cd = 0.5;

% Initialization
v0 = 10; % Initial velocity
th0 = pi/3; %Initial launch angle

% Populating the initial state vector
Y0 = [0; v0*cos(th0); 0; v0*sin(th0)];

% Time span
tSpan = linspace(0,10,10^3);

int0 = [v0; th0];

% Optimization: Minimize the launch velocity such with an
appropriate launch angle
% Objective: x(tf) = 2 m
options = optimoptions('fmincon','stepTolerance',1e-
12,'FunctionTolerance',1e-12,...
    'OptimalityTolerance',1e-
12,'ConstraintTolerance',1e-12);
int =
fmincon(@ode_22rb,int0,[],[],[],[],[0,0],[20,pi/2],[],optio
ns);

% Integration
opts.relTol = 1e-12;
opts.absTol = 1e-12;
opts.Events = @zeroHeight;

% Re-initialize with optimized parameters
Y0_opt = [0; int(1)*cos(int(2)); 0; int(1)*sin(int(2))];

```

```

% Integration
[T,Y] = ode45(@ode_22a,tSpan,Y0_opt,opts,p);

% Graphics
figure(1)
plot(Y(:,1),Y(:,3),'b',Y(1,1),Y(1,3),'k*',Y(end,1),Y(end,3)
,'ko')
title('Cannon Ball w/ Quadratic Drag');
xlabel('x(m)');
ylabel('y(m)');

fprintf('The minimum velocity is: %3.3e m/s \n',int(1));
fprintf('The required angle is: %3.3e rad \n',int(2));

function dydt = ode_22a(t,y,p)

vmag = sqrt(y(2)^2+y(4)^2);

dydt(1) = y(2);
dydt(2) = -p.cd/p.m*y(2)*vmag;
dydt(3) = y(4);
dydt(4) = -p.g-p.cd/p.m*y(4)*vmag;

dydt = dydt';

end

function F = ode_22rb(int0)

% System parameters
p.m = 1;
p.g = 10;
p.cd = 0.5;

% Populating the initial state vector
Y0 = [0; int0(1)*cos(int0(2)); 0; int0(1)*sin(int0(2))];

% Time Span
tSpan = linspace(0,10,10^3);

opts.relTol = 1e-12;
opts.absTol = 1e-12;

```

```
opts.Events = @zeroHeight;
```

```
[T,Y] = ode45(@ode_22a,tSpan,Y0,opts,p);
```

```
F = abs(2-Y(end,1));
```

```
End
```

```
function [value,isterminal,direction] = zeroHeight(t,y,p)
```

```
% This function creates an event for the time when the  
cannon ball hits
```

```
% the ground
```

```
value = y(3); % Trigger when height gets to 0.
```

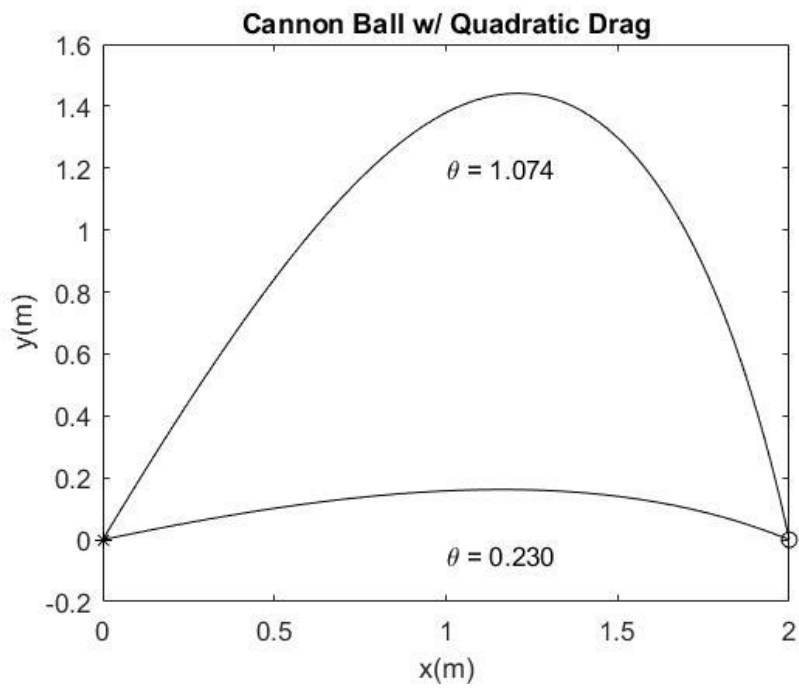
```
isterminal = true; % Stop integrating when this happens.
```

```
direction = -1; % Trigger when going negative (not  
positive)
```

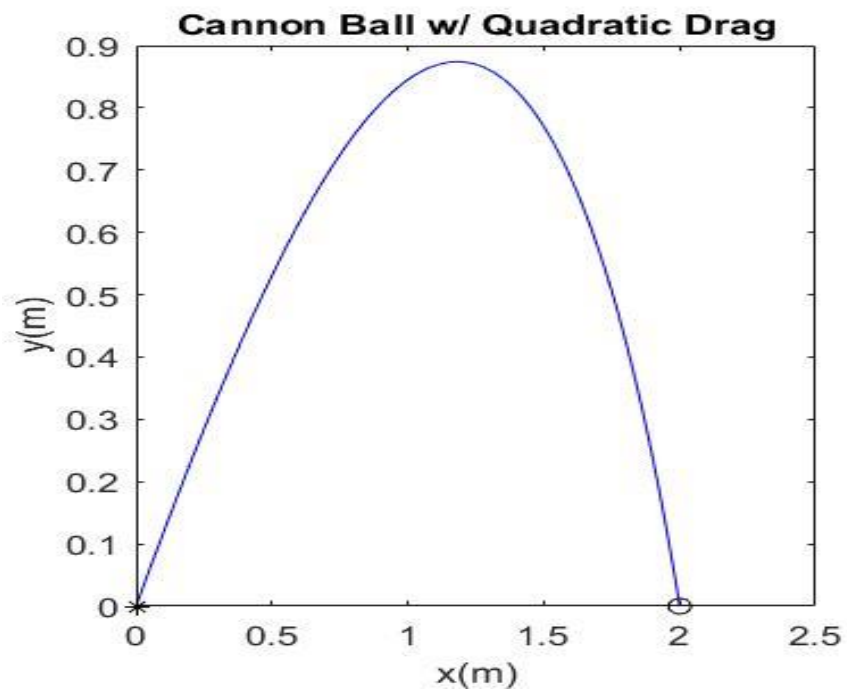
```
end
```


Plots:

Part a:



Part b:



Comparing the unoptimized velocity with the optimization velocity:

